

Newton Divided Difference form

Polynomial

Expression: $P_n(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1) + a_3(x-x_0)(x-x_1)(x-x_2) + \dots + a_n(x-x_0)\dots(x-x_{n-1})$

Now

$$a_0 = f[x_0]$$

$$a_1 = f[x_0, x_1]$$

$$a_2 = f[x_0, x_1, x_2]$$

$$\left. \begin{array}{l} a_0 = f[x_0] \\ a_1 = f[x_0, x_1] \\ a_2 = f[x_0, x_1, x_2] \end{array} \right\} a_n = f[x_0, x_1, x_2, \dots, x_n]$$

plugging $a_0, a_1, a_2, \dots$ into polynomials.

$$P_n(x) = f[x_0] + f[x_0, x_1](x-x_0) + f[x_0, x_1, x_2](x-x_0)(x-x_1) + \dots + f[x_0, x_1, \dots, x_n](x-x_0)\dots(x-x_{n-1})$$

Sample Example

x	$f(x)$
0	10 ----- x_0
1	20 ----- x_1
2	30 ----- x_2

$[2] \Rightarrow$ Degree, so $P_2(x)$ needed

$$P_2(x) = f[x_0] + f[x_0, x_1](x-x_0) + f[x_0, x_1, x_2](x-x_0)(x-x_1)$$

$$x_0 = 0 \quad f[x_0] = 10$$

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0} = \frac{20 - 10}{1 - 0} = 10$$

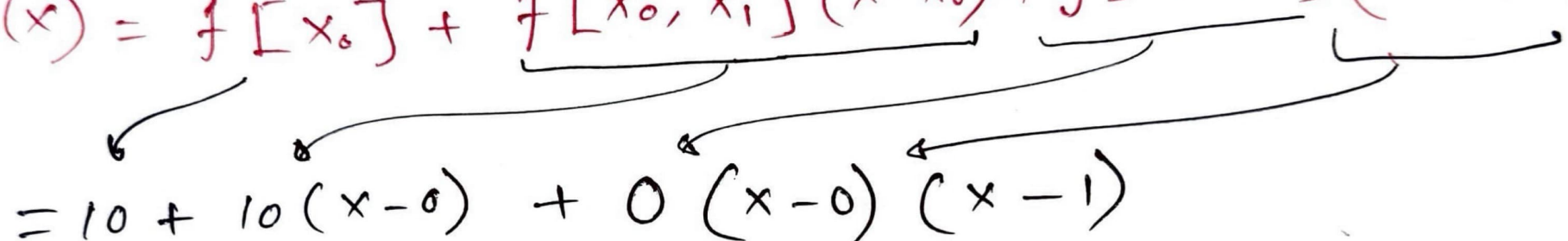
$$x_1 = 1 \quad f[x_1] = 20$$

$$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1} = \frac{30 - 20}{2 - 1} = 10$$

$$x_2 = 2 \quad f[x_2] = 30$$

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = \frac{10 - 10}{2 - 0} = 0$$

plugging all values

$$P_2(x) = f[x_0] + f[x_0, x_1](x-x_0) + f[x_0, x_1, x_2](x-x_0)(x-x_1)$$

$$= 10 + 10(x-0) + 0(x-0)(x-1)$$

Say. $P_2(5) = 10 + 10(5-0) + 0 \cdot (5-0)(5-1)$
 $= \text{ANS.}$

a new value added:

x	$f(x)$
3	40 --- x_3

$$x_0 = 0 \quad f[x_0] = 10$$

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0} = \frac{20 - 10}{1 - 0} = 10$$

$$x_1 = 1 \quad f[x_1] = 20$$

$$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1} = \frac{30 - 20}{2 - 1} = 10$$

$$x_2 = 2 \quad f[x_2] = 30$$

$$f[x_2, x_3] = \frac{f[x_3] - f[x_2]}{x_3 - x_2} = \frac{40 - 30}{3 - 2} = 10$$

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = \frac{10 - 10}{2 - 0} = 0$$

$$f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1} = \frac{10 - 10}{3 - 1} = 0$$

$$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0} = \frac{0 - 0}{3 - 0} = 0$$

$$x_3 = 3 \quad f[x_3] = 40$$

a new value added:

x	$f(x)$
3	40 --- x_3